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Effective Motion Modeling for UAV-platform Multiple Object Tracking with Re-Margin Loss

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Abstract

Multiple object tracking (MOT) from unmanned aerial vehicle (UAV) platforms requires efficient motion modeling. This is because UAV-MOT faces tracking difficulties caused by large and irregular motion, and insufficient training due to the motion long-tailed distribution of current UAV-MOT datasets. Previous UAV-MOT methods either extract motion and detection features redundantly or supervise motion model in a sparse scheme, which limited their tracking performance and speed. To this end, we propose a flowing-by-detection module to realize accurate motion modeling with a minimum cost. Focusing on the motion long-tailed problem that were ignored by previous works, the flow-guided margin loss is designed to enable more complete training of large moving objects. Experiments on two widely open-source datasets show that our proposed model can successfully track objects with large and irregular motion and outperform existing state-of-the-art methods in UAV-MOT tasks.

1. Introduction

Multiple object tracking (MOT) aims to identify objects in a given video and is widely used in computer vision [8, 20, 35], including applications in autonomous driving [16], human-computer interaction [7], and pedestrian tracking [36, 40]. A common MOT method consists of two main stages: detection and association [44]. The detection stage identifies all objects in each frame, while the association stage matches objects across consecutive frames to establish complete trajectories for each object [34]. Recently, MOT in unmanned aerial vehicle (UAV) platforms has attracted extensive research interest. Compared with conventional MOT [9, 26], MOT in UAV view faces more challenges.

Firstly, both the ground object and the UAV platform have fast and irregular motion, posing a challenge for the tracker to accurately follow the object. Secondly, the fast and irregular motion will reduce the image quality [25] and affect the detection of the object. Thirdly, existing UAV-MOT datasets face motion long-tailed problems, with larger motion samples having lower ratios in the dataset, which is different from conventional long-tailed problem [50] that focus on class-imbalances. Previous study [54] defines relative velocity (Eq (1)) to reveal the large and irregular motion

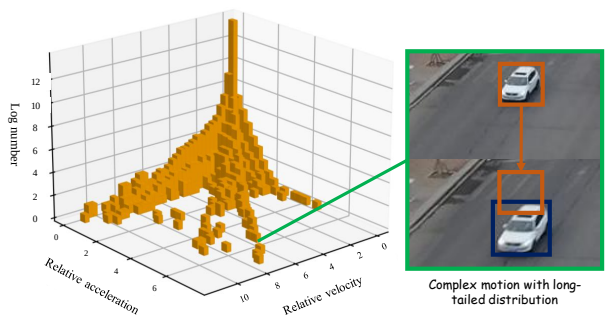

Figure 1. The 3D histogram of Visdrone dataset shows a motion long-tail distribution in UAV-MOT scene. Each bar indicates a log number of samples with given relative acceleration (RA) and relative velocity (RV). The larger the RA and RV, the less the training sample number, which leads to insufficient training of MOT model in those difficult-to-track situations.

problems. We further define relative acceleration (Eq (2)) and plot 3-D histogram of current UAV-MOT dataset (as Fig. 1 shows), which reveals that current UAV-MOT dataset faces motion long-tailed problems, with few objects having higher relative velocity and relative acceleration than the remaining objects. This will lead to insufficient training of our tracker in videos with those challenging situations, which will consequently limit the tracking performance.

For MOT in large and irregular motion scenes, previous studies have attempted to solve it by using motion modeling [44, 54, 59]. Kalman filtering and its various variants are very commonly used motion modeling schemes [2,5,33, 55]. However, the kalman filtering scheme is a non-learning algorithm, which can only make pre-assumptions about the motion (e.g., assume linear motion), therefore, the accuracy is limited in large and irregular motion scenes. Some scholars have proposed to use neural networks to learn the motion of an object [44, 59], and some scholars have directly extracted the optical flow to represent the motion of an object [54]. These methods have achieved some acceptable results, but they also suffer from the problem of sparse supervision or excessive computational overhead, which restricts the algorithm performance and speed. In addition, previous studies have ignored the the motion long-tailed distributions on UAV-MOT datasets, which makes difficult-to-detect objects (with large motion) less trained compared with easy-to-detect objects (with small motion).

To address these issues, we propose FM-Tracker for fast and accurate motion modeling and motion-long-tail-focused object detector training. Unlike previous work that redundantly extracts object detection features and motion features from raw input images, FM-Tracker extracts optical flow information based on features extracted by the object detector, which greatly reduces the computational effort of motion modeling, outperforms previous motion modeling methods in large and irregular motion scenes. In addition, FM-Tracker improves the detection performance in large and irregular motion scenes by proposing an optical flow-guided marginal function to impose a larger classification loss for samples with larger motions, thus effectively mitigating the problem of motion long-tailed distribution.

The main contributions of this paper are summarized as follows:

- We propose the flowing-by-detection module (FMD), which modeling object motion by estimating optical flow from detection features and reached fast and accurate motion modeling.
- We propose the flow-guided margin loss (FGMLoss), which use optical flow to supervise the training of the object detector, and improves the detection performance on fast-moving objects.
- The FM-Tracker proposed based on flowing-by-detection module and flow-guided margin loss reached state-of-the-art on two public MOT-in-UAV-view datasets, which promoted the progress of multiple object tracking in UAV scenes.

2. Related Works

Existing MOT methods typically follow the detection and association scheme [1, 13, 35, 38, 41, 42], with the detection step detects the object in the current frame, and the association step associates the previous object trajectory with the currently detected object based on a specific criterion. The association criterion can be classified into two types: positional information [14, 23, 48, 57] and appearance information [24, 28, 39, 45, 53]. Association based on positional information needs to model the object motion to predict the position of the object at the next moment [2, 3, 44, 55, 59], while methods based on appearance information need to extract the appearance information of the object to compute the appearance similarity of the object between frames [51, 52, 56, 58]. The small object size, blurred object appearance, and very similar appearance between ground vehicles makes appearance information unreliable for object association in UAV-MOT scenes [54]. Therefore, this work focuses on motion modeling as association.

2.1. Motion Modeling in MOT

The most commonly used motion modeling methods in MOT are Kalman filtering (KF) [21] and its various variants [2, 5, 55]. These methods first make assumptions about the motion patterns of the objects (e.g., assume linear motion) and maintain a KF tracker for each object. The KF tracker then use the object position detected by the detector as observation information to continuously update the

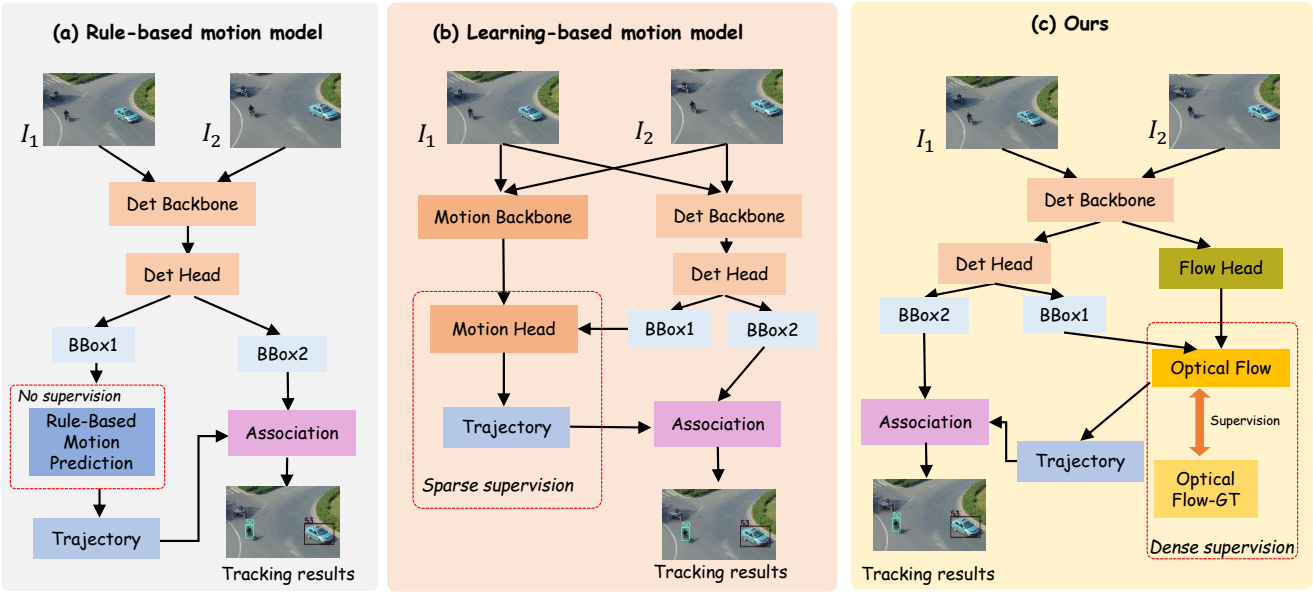

Figure 2. Differences between our proposed method and other motion modeling methods. (a) Rule-based motion modeling uses pre-set rules with no supervisory information, making it difficult to adapt to complex motion. (b)Previous learning-based motion modeling repeats feature extraction while the supervisory information is sparse for object offsets. (c) Our method modeling motion from detector features and using high-precision optical flow for dense supervision, which is faster and more accurate.

intrinsic distribution of the tracker. The updated distribution are used to predict the position of the object in the next frame. These methods require prior assumptions about object motion patterns and their probability distribution without any learnable parameters, thus limiting their performance in scenes with large and irregular motions. Some scholars use neural networks to learn the motion of objects, for example, CenterTrack [59] uses an additional CNN to predict object offsets, SiamMOT [44] uses a motion modeling network to predict object offsets, and FOLT [54] proposes to use a lightweight optical flow network [22] to extract the optical flow and use the optical flow to predict the object motion. These methods introduce learnable parameters to improve the accuracy. However, they either performed repetitive feature extraction or used only sparse supervised information, which brings additional computational overhead and limits the real-time performance of the MOT algorithm. In this paper, we propose flowing-by-detection module to predict optical flow from detection features, which leads to a significant reduction in computational overhead and achieves accurate and fast motion modeling.

2.2. Long-Tailed Classification

The conventional long-tailed problem [50] refers to the imbalance between different classes that multiple classes only account for a small proportion of the dataset, while a few classes account for a large proportion of the dataset. This imbalance results in classes with few training samples not being adequately trained, affecting the classification accuracy. Many scholars have proposed solutions to this problem, such as resampling [18,30,43], Loss reweighting [27, 29, 31], re-margining [6, 37, 49]. In this paper, we point out that the UAV-MOT task faces the motion long-tailed distribution problem by defining relative velocity and relative acceleration, with few objects having higher relative velocity and relative acceleration than the remaining objects. Different from conventional class long-tailed problem, the problem of motion long-tailed distribution is still under investigation. To this end, we propose flow-guided margin loss, which measures the motion complexity of an object by optical flow and imposes a classification loss function with larger boundaries for objects with larger motions, thus better solving the problem of less training for large motion objects.

2.3. Optical Flow Estimation

The optical flow estimation takes two adjacent images as input and predicts the offset of each pixel in the next image, which is suitable for modeling the motion of the object. Traditional optical flow methods [17, 46] are computationally intensive and can only predict sparse optical flow, therefore unsuitable for application to motion modeling in MOT tasks. Recently, deep neural networks have been widely used in the field of optical flow estimation [11, 19, 47]. However, the computational overhead of these networks is still huge, and most of them cannot reach real-time. Some lightweight methods [22] are capable of real-time, but not

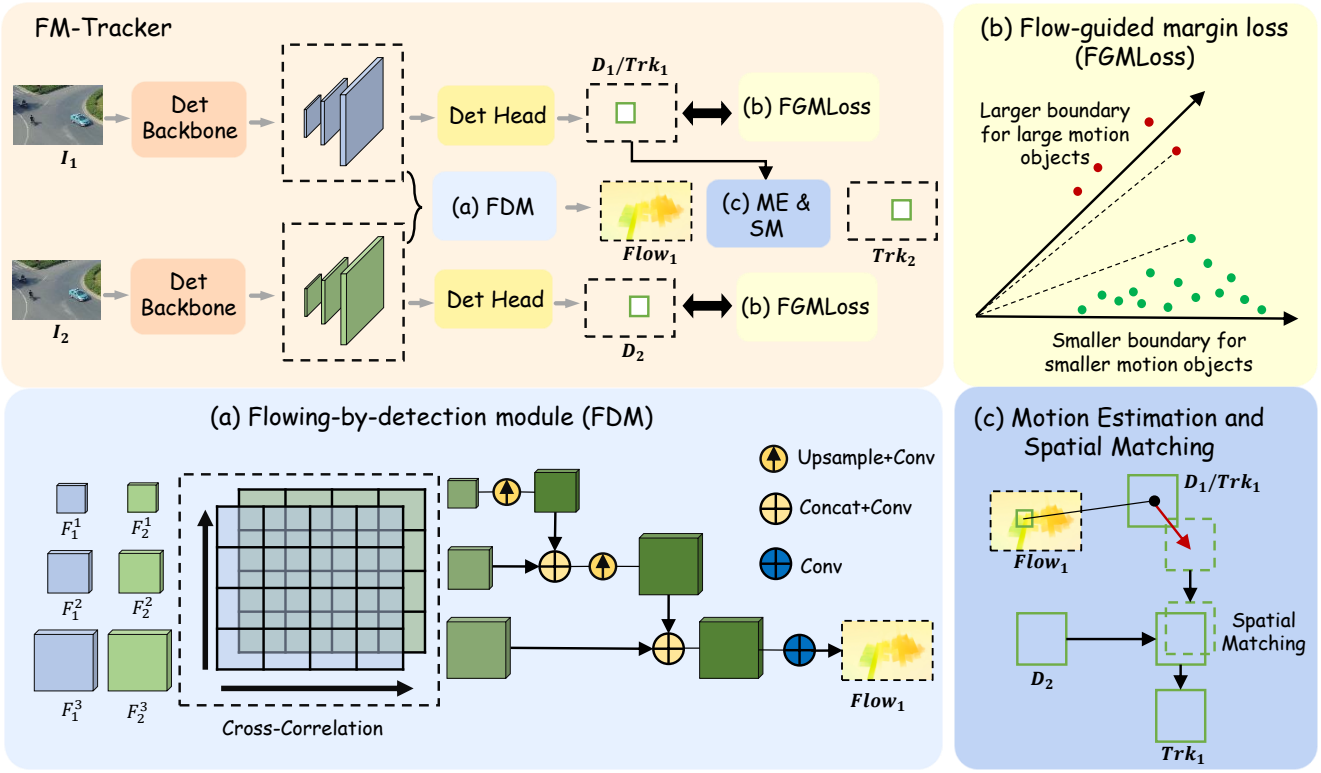

Figure 3. Overall structure of our method. Multi-scale detection features are first extracted using a detection backbone, which is fed into the detection head to output the object bounding box, score, and category. The object score is optimized using the proposed FGMLoss. The detection feature is also fed into the proposed flowing-by-detection module (FDM), which captures the difference between the two detection features and predicts the optical flow, which is supervised by the L1-Loss and the output of the large optical flow model. Afterward, the position of the object in the previous frame in the next frame is predicted based on the optical flow value, and the predicted position of the object is matched with the detected position in the current frame to obtain a new object trajectory.

as accurate as larger networks. To this end, we propose flowing-by-detection module (FDM) to estimate the optical flow by extracting the difference in detection features between adjacent frames, which effectively reuses the features of the detector. In addition, we supervise our FDM using a pre-trained large optical flow network [10], which achieves better tracking accuracy and maintains the tracking speed in real-time.

3. Proposed Method

3.1. Overview

Different from conventional MOT, UAV-MOT faces larger and more irregular motions and motion long-tailed problems, which is revealed by relative velocity [54] and relative acceleration:

$$RV(t) = \frac{\sqrt{(cx_t - cx_{t-1}))^2 + (cy_t - cy_{t-1}))^2}}{\sqrt{w_t^2 + h_t^2}}, \quad (1)$$

where cx_t, cy_t denote the object center position at time t, w_t, h_t denote the object width and height, respectively at time t.

$$RA(t) = RV(t) - RV(t-1), \quad (2)$$

Fig. 1 illustrates the motion long-tailed problem in existing UAV-MOT datasets. Here we propose a flowing-by-detection module (FDM) to modeling complex motion in a minimum cost. The FDM first extracts the multi-scale difference information from the current detection features, which is the key of low-latency (4.1ms). Then the extracted features are upsampled and fused from the bottom to top, scale-by-scale. Finally, the fused difference features are convolved to output an 2-channel feature map, which is supervised by large optical flow networks to reach high-accuracy motion modeling. Aiming at the motion long-tailed problem of UAV-MOT, we propose flow-guided margin loss (FGMLoss). The FGMLoss calculates the classification boundaries based on the optical flow values of the objects, and assigns larger classification boundaries for objects with larger motions. The larger boundary forcing the detector to focus more on objects with larger motions, and improving the detection of large-motion objects efficiently.

We adopt YOLOX-S [15] as our base object detector and

use FGMLoss to supervise the detector. Then the detector weights are frozen, and the 3-scale FPNs output from the YOLOX-FPN network are used as inputs to the FDM, which estimates optical flow for the current frame and predicts object motion according to the flow. We use the optical flow output of the latest high-precision optical flow extractor EMD-Flow [10] to supervise the FDM.

The optical flow is estimated by FMD using cached FPNs from previous frame and currently extracted FPNs. The object positions are updated according to its optical flow at centroid. Finally, the updated object positions are spatially matched with current detected object positions, which generate object trajectory at current time.

3.2. Flowing-by-Detection Module

The flowing-by-detection module (FDM) extracts difference information from existing multi-scale detection features and uses this information to predict optical flow, enabling fast and highly accurate motion modeling. As shown in Fig. 3(a), the FDM takes two FPN features as input (denoted as (F_1^1, F_1^2, F_1^3) and (F_2^1, F_2^2, F_2^3)). Each FPN feature contains three scales, and the size of each scale is (1/8,1/16,1/32) of the original image size, respectively. The FDM first uses a cross-correlation layer for each pair of feature maps to extract neighborhood correlations between pairs. Then, starting from the smallest scale, the correlation map is bi-linearly upsampled and convolved, and the upsampled correlation map is channel-concatenated with the correlation map of the upper scale and then convolved to achieve effective fusion. The feature map of the top scale is convolved to output an optical flow estimation with 2 channels, where the first channel represents the x-direction offset of the current position and the second channel represents the y-direction offset of the current position. Each convolution layer consists of a 3x3 convolution with a batch normalization layer and a softshrink activation function [32]. The output optical flow maps are supervised using high-precision optical flow output from EMD-Flow, which is the key to achieve high accuracy motion modeling. The objective function between FDM-estimated optical flow and EMD-Flow estimated one is L1-Loss.

After the optical flow estimation, we estimate every object's motion dx, dy according to its optical flow value at object centroid cx, cy:

$$\begin{aligned} dx &= flow_x(cx, cy), \\ dy &= flow_y(cx, cy), \end{aligned} \tag{3}$$

where $flow_x, flow_y$ are the x and y component of optical flow estimation, respectively.

3.3. Flow-Guided Margin Loss

Fig. 1 illustrates the motion long-tail problem in existing UAV-MOT datasets, resulting in insufficient training of our tracker in the most complex motion scenes. For the object tracking task, a small number of large-scale motion scenes greatly affects the completeness of the tracking trajectory because in some cases the interrupted trajectory is difficult to track back. Therefore, it is important for our tracking algorithm to focus more on the object with large motion in the training stage. Inspired by the LDAM loss function [6] to assign different decision boundaries to different categories of objects, we propose flow-guided margin loss to assign different decision boundaries to objects with different optical flow counted in its bounding box. Specifically, we calculate the optical flow boundary of each target frame based on the mean magnitude of the optical flow:

$$M_i = s * (\sigma(M_{flow}(B_i)) - 0.5), \tag{4}$$

where s is the amplification factor, the default setting is 10, the $M_{flow}(B_i)$ is defined by Eq. (5):

$$\begin{aligned} M_{flow}(B_i) &= \frac{1}{3 \times 3} \\ &\sum_{j=c_x-1}^{c_x+1} \sum_{k=c_y-1}^{c_y+1} \sqrt{Flow^2(j,k,x) + Flow^2(j,k,y)}, \end{aligned} \tag{5}$$

where B_i denote the i-th object, (c_x,c_y) denote its center coordinates. Sigmoid function σ is defined as:

$$\sigma(x) = \frac{1}{1 + e^{-x}}. \tag{6}$$

For the classification of each object box, the flow-guided margin loss is defined as:

$$\begin{aligned} \text{FGMLoss}(y_i, \hat{y}_i, D_i) = -y_i * \log(\sigma(\hat{y}_i - D_i)) - \\ (1 - y_i) * \log(1 - \sigma(\hat{y}_i)), \end{aligned} \tag{7}$$

where $\hat{y}_i$ is the predicted class probability, y_i is the ground-truth label. The function of subtracting D_i from the output of the network classification layer is to assign different decision boundaries to different object boxes based on their optical flow value. The value of D_i subtracted from the box with a large optical flow value is larger, thus forcing the model to output a higher probability for the target box with a large range of motion.

4. Experiments

4.1. Datasets and Metrics

Datasets.

We conduct comparative experiments on the Visdrone [60] and UAVDT [12] datasets. These two datasets are open source multi-class multi-object tracking datasets, and both are collected from the perspective of UAVs. Therefore, it is suitable to study the tracking objects with small sizes and irregular motion problems in these two datasets.

The Visdrone dataset consists of a training set (56 sequences), validation set (7 sequences), test-dev set (17 sequences), and test-challenge set (16 sequences). There are 10 categories in the Visdrone dataset: pedestrian, person, car, van, bus, truck, motor, bicycle, awning-tricycle, and tricycle. Each object within the above categories is annotated by a bounding box, category number, and unique identification number. In experiments of Visdrone, we use the full ten categories in training while only using five categories in testing, i.e., car, bus, truck, pedestrian, and van in evaluation, as the evaluation toolkit offered by Visdrone officials only evaluates these five categories.

The UAVDT dataset is a car-tracking dataset based on aerial view. It includes different common scenes, such as squares, arterial streets, and toll stations. UAVDT dataset consists of a training set (30 sequences), and a test set (20 sequences), with three categories: car, truck, and bus. In experiments of UAVDT, all three categories are evaluated using Visdrone's official evaluation toolkits.

Metrics. We select MOTA and IDF1 as our main evaluation metrics. The MOTA is calculated as:

$$MOTA = 1 - \frac{FP + FN + IDs}{GT}, \tag{8}$$

where FP denotes numbers of false positives, FN denotes numbers of false negatives, IDs denote numbers of ID switches, and GT denotes numbers of ground-truth objects. The IDF1 is calculated as:

$$IDF1 = \frac{2IDTP}{2IDTP + IDFP + IDFN}, \tag{9}$$

where IDTP, IDFP, and IDFN denote numbers of true positive, false positive, and false negative that consider ID information.

Table 1. Effectiveness of flowing-by-detection module. OF-Net denote method that directly using Optical Flow Network to inference, which is slower than our FDM. The best result is marked in bold. The $'\uparrow'$ ($'\downarrow'$) means that the higher (lower) result is better.

Motion modeling	Flow Network	MOTA↑	IDF1↑	Time(ms)↓
KF	-	39.6	50.4	15.2
OF-Net	FastFlowNet	40.4	53.1	**11**
OF-Net	GMA	40.5	53.5	102
OF-Net	EMD-Flow	**40.7**	**54.0**	115
FDM (Ours)	FastFlowNet	40.2	52.5	4.1
FDM (Ours)	GMA	40.4	52.8	4.1
FDM (Ours)	EMD-Flow	**40.5**	**53.0**	**4.1**

4.2. Implementation Details

In all experiments, we keep the same train-test split as the official split of the Visdrone and UAVDT datasets. We use the YOLOX-S [15] model as the base object detector on both datasets, the input image size is 1088×608. We use the stochastic gradient descent method [4] to optimize the detector, the learning rate is set to 0.0001, the batch size is set to 8, each dataset is trained for 10 epochs, and the training and testing are completed on a single 2080TI graphic card. The flowing-by-detection module and the object regression branch of the detector are trained using the L1 loss function. The object classification branch of the detector is trained using the proposed FGMLoss. We select the accuracy evaluation tool officially provided by the Visdrone dataset to complete the comparison of all metrics.

Table 2. Comparisons of different imbalance training loss functions in Visdrone test-dev dataset. The $'\uparrow'$ means that the higher result is better. The best result is marked in bold.

Loss function	MOTA↑ (%)	IDF1↑ (%)
Cross Entropy	39.6	50.4
Focal	36.4	48.3
LDAM	40.3	51.0
FGML (Ours)	**42.1**	**54.4**

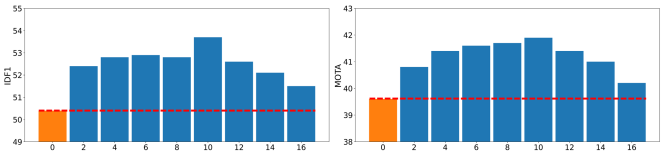

Figure 4. Comparisons of flow-guided margin loss with different s values. The orange bar denote accuracy of baseline model, the blue bar denote accuracy of model with different FGMLoss values. The red dash line shows that no matter what s value choose, the FGMLoss always performs better than baseline model.

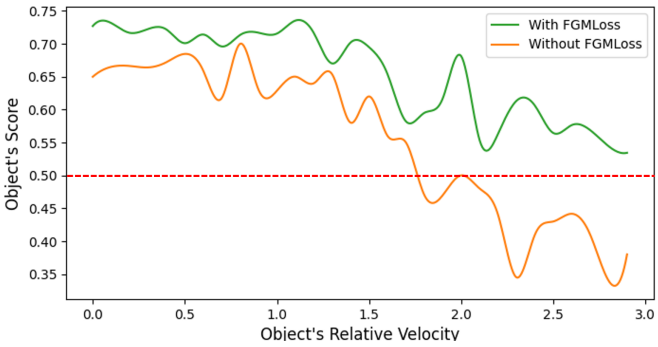

Figure 5. Visualization of object's detection scores with different training loss. With FGMLoss introduced, the average detection score of large motion objects are higher than 0.5 (denoted as red dash line), which is more reliable than model without FGMLoss.

4.3. Ablation study

Baseline model. The baseline model we compared with is the model that uses YOLOX-S as detector, extended Kalman Filter as motion modeling [5], and two-stage spatial matching as association [55]. **Different optical-flow**

Table 3. Ablation studies on Visdrone and UAVDT test set. The $'\uparrow'$ ($'\downarrow'$) means that the higher (lower) result is better. The best result is marked in bold.

			Visdrone		UAVDT		
Baseline	FGML	FDM	MOTA(%) ↑	IDF1(%) ↑	MOTA(%) ↑	IDF1(%) ↑	Time(ms)↓
√			39.6	50.4	47.5	63.1	15.0
√	√		42.1	54.4	49.2	64.1	15.0
√		√	40.5	53.1	49.7	66.1	4.1
√	√	√	**44.1**	**57.5**	**51.4**	**68.9**	**4.1**

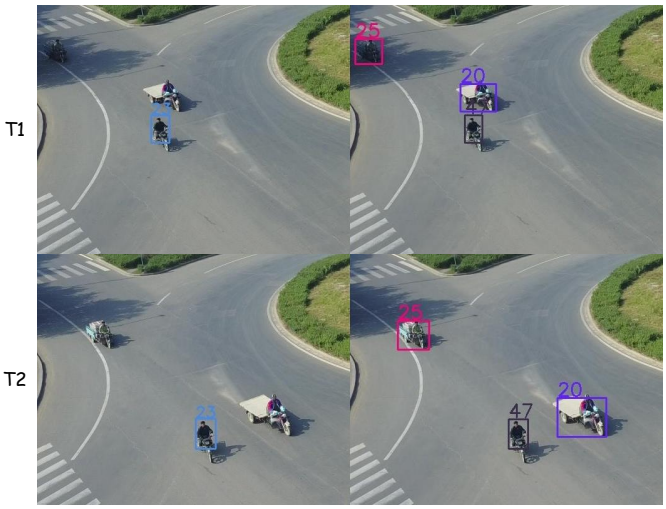

Figure 6. Visualization results for large and irregular motion scenes. The model without FDM missed the fast-moving tricycles on the road, while the model with FDM tracked them with ID 20, 25, and 47.

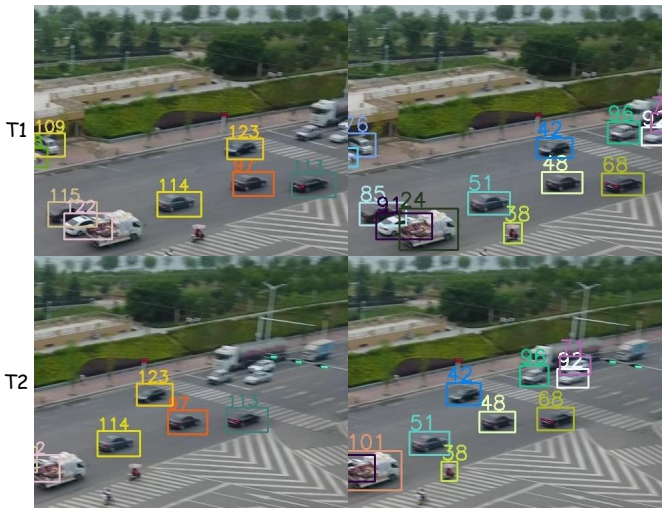

Figure 7. Visualization results for blurred motion scenes. The model without FGML missed the fast-moving bikes on the crossroad while the model with FGML tracked it with ID 38.

extractor. We compare the performance of different optical flow networks used for tracking and also the performance of FDM trained using different optical flow networks. Table 1 shows that the tracking accuracy obtained using larger optical flow networks is higher. For example, FastFlowNet, GMA, EMD-Flow reach 53.1, 53.5, and 54.0 in IDF1, respectively. However, larger optical flow networks cost more inference time (11, 102, and 115 ms, respectively). By using the proposed FDM, the additional overhead is greatly reduced and the inference time is much lower (4.1ms vs 115ms). Meanwhile, using the optical flow network with higher accuracy to supervise the FDM can achieve better results (52.5, 52.8, 53.0 in IDF1, respectively) without increasing the inference time. This allows our method to improve together with the improvement of large optical flow networks.

Flow-guided margin loss. We compare the accuracy of the proposed flow-guided margin loss (FGMLoss) with other long-tailed loss functions. Table 2 shows that the FGMLoss proposed in this paper performs better than the previous category long-tailed loss functions (MOTA reaches 42.1, compared to 40.3 36.4, and IDF1 reaches 54.4, compared to 51.0 and 48.3), which proves the effectiveness of our proposed FGMLoss.

Fig. 7 visualized our ablation results. As shown in Fig. 7, the model using FGMLoss successfully tracked the moving bike-rider with blurred appearance (denote as ID 38), while the baseline model failed to track it. The qualitative and quantitative results above confirm the effectiveness of our proposed loss function.

We also evaluated the importance of the hyperparameter s from flow-guided margin loss. We set s to 2, 4, 6, 8, 10, 12, 14, and 16, and tested their accuracy on the multiple object tracking task. As Fig. 4 shows, our flow-guided margin loss is always better than the original cross-entropy loss regardless of the setting of s value. Fig. 4 also shows that the best results on Visdrone test set are achieved when s is set to 10, with MOTA reaching 42.1% and IDF1 reaching 54.4%, respectively. Therefore, we keep s=10 fixed in subsequent experiments.

We further investigate the effectiveness of FGMLoss to

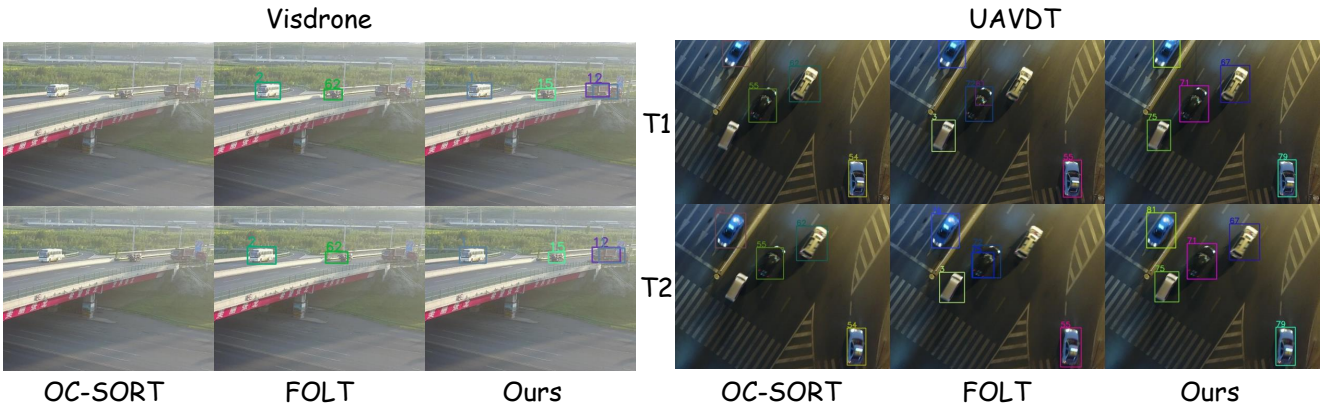

Figure 8. Visual comparisons on the Visdrone and UAVDT datasets. The same number denotes the same object in different frames.

improve the detection of fast-moving objects. As Fig. 5 shows, introducing FGMLoss effectively increases the average object scores of large motion objects (objects with larger relative velocity). When using FGMLoss, the average score of large motion objects are higher than 0.5, which is more reliable than model without FGMLoss.

Table 4. Comparisons of the proposed FM-Tracker with state-of-the-art methods on Visdrone and UAVDT test sets. The best result is marked in bold. The $'\uparrow'$($'\downarrow'$) means that the higher (lower) result is better. Our proposed FM-Tracker surpasses state-of-the-art in both tracking accuracy and inference speed.

	Visdrone		UAVDT		
Method	MOTA↑	IDF1↑	MOTA↑	IDF1↑	FPS↑
SORT [2]	14	38	39	43.7	23.5
IOUT [3]	28.1	38.9	36.6	23.7	27.3
SiamMOT [44]	31.9	48.3	39.4	61.4	11.2
FairMOT [56]	34.3	46.1	41.5	59.2	17.2
ByteTrack [55]	35.7	37	41.6	59.1	27.0
UAVMOT [34]	36.1	51	46.4	67.3	12.0
OC-SORT [5]	39.6	50.4	47.5	64.9	26.4
FOLT [54]	42.1	56.9	48.5	68.3	29.4
U2MOT [33]	42.8	53.9	47.1	65.2	24.1
Ours	**44.1**	**57.5**	**51.4**	**68.9**	**34.1**

Flowing-by-detection module. We compare FDM with the OC-SORT tracker, and direct motion prediction using the optical flow network. As shown in Table 1, the methods of motion modeling using optical flow are both better than the extended Kalman filtering used by OC-SORT. Compared to predicting motion directly using the optical flow network, the FDM is comparable in accuracy (40.4 vs 40.5 in MOTA, 53.1 vs 53.0 in IDF1), while the inference time is significantly lower than using the optical flow network directly (10.3 vs 4.1 in ms). As Fig. 6 shows, after using our FDM, the tracker successfully tracks the three fast-moving tricycle on the road (with ID1 20, 25, 47), while the model without FDM only tracked one tricycle (with ID 23). In summary, the qualitative and quantitative results above confirm the effectiveness of our proposed FDM in improving object tracking with large and irregular motion.

Combination. We also evaluate the effectiveness of our FGML and FDM which are used in combination. As Table 3 shows, both FDM and FGML can effectively improve the tracking performance on Visdrone dataset (39.6% to 40.5% and 39.6% to 42.1% in MOTA, 50.4% to 53.1% and 50.4% to 54.4% in IDF1) and UAVDT dataset (47.5% to 49.7% and 47.5% to 49.2% in MOTA, 63.1% to 66.1% and 63.1% to64.1% in IDF1). With the FGML and FDM commonly used, the best tracking performance is obtained (44.1% in MOTA and 57.5% in IDF1 on Visdrone, 51.4% in MOTA and 68.9% in IDF1 on UAVDT), and the motion-inference time is lower than the baseline model (15.1 to 4.1 in ms).

4.4. Comparison with state-of-the-art

As shown in Table 4, the FM-Tracker proposed in this paper exceeds the current state-of-the-art (SOTA) methods in several important metrics such as MOTA, IDF1. For example, the MOTA of FM-Tracker is 1.3% higher than the previous highest method FOLT (44.1% compared with 42.8%), while the inference speed is also faster than the previous fastest method FOLT (34.1 compared with 29.4). FM-Tracker also achieves the best MOTA and IDF1 in the UAVDT dataset, with MOTA 2.9% higher than the previous best method FOLT (51.4% compared with 48.5%) showing the strongest ability in tracking objects from a UAV view.

We also visualize the comparison results of the latest best methods (OC-SORT and FOLT) and our FM-Tracker in the Visdrone dataset and the UAVDT dataset. As the left part of Fig. 8 shows, our FM-Tracker successfully tracks the small and fast-moving bus and truck (with object ID 1, 15, and 12) in both $T1$ and $T2$ frames, while the OC-SORT failed to track these three vehicles, and the FOLT failed to track the half-occluded truck with ID 12. Therefore, our FM-Tracker is better than FOLT and OC-SORT in tracking objects with small sizes and fast motion. As the right part of Fig. 8 shows, the camera and the object are both in motion, introducing large and irregular motion and causing the

object feature to blur. In this difficult situation, our FM-Tracker successfully tracks all five cars in frame $T1$ with object ID 67, 71, 75, 79, and 81, respectively, while the OC-SORT missed the left-bottom car and the FOLT has duplicated tracking of the same car with ID 72,61 at T1 time and 72,75 at T2 time. The FOLT also missed the right-top car which is severely blurred due to large and irregular motion. In summary, our FM-Tracker performs better than the state-of-the-art in objects with small sizes, large and irregular motion, and blurred appearance.

5. Conclusion

In this paper, we propose FM-Tracker to address challenging problems of UAV-MOT. Focusing on the problem that current motion modeling trackers are sparsely supervised or have duplicated feature extraction on detection and motion modeling, we propose flow-by-detection module (FDM). The FDM extracts optical flow information based on features extracted by the object detector, which greatly reduces the computational cost of motion modeling, and outperforms previous motion modeling methods in large and irregular motion scenarios. Aiming at addressing the motion long-tailed problems in current UAV-MOT datasets, we propose flow-guided margin loss (FGML), which uses optical flow to supervise the training of the object detector, and improves the detection performance on fast-moving objects. Experiments show that both FGML and FDM can improve the accuracy of multiple object tracking in UAV view. Comparison with state-of-the-art shows that our FM-Tracker achieves the best results on the two public UAV-MOT datasets in both tracking accuracy and inference speed.

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